MIL-P-45455E (AR)
7 November 1979
SUPERSEDING
MIL-P-45455D
29 December 1970

MILITARY SPECIFICATION

PROJECTILE, 175MM, HE, M437 LOADING, ASSEMBLING AND PACKING

This specification is approved for use by US Army Armament Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the loading, assembling and packing for the Projectile, 175MM, HE, M437.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification For

STANDARDS

MILITARY

MIL-STD-453 - Inspection, Radiographic
MIL-STD-105 - Sampling Procedures and Tables For Inspection by Attributes

FSC: 1320

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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DRAWINGS (see 6.9)

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

8837902 - Projectile, 175MM, HE, M437A1 and M437A2, for 175MM Gun M113
8857344 - Pallet for Projectile, 175MM

3. REQUIREMENTS

3.1 Materials. Materials and parts shall be in accordance with the applicable drawings and specifications.

3.1.1 Comp B - Clean explosive scrap. Riser scrap shall be blended with virgin explosive only in an amount such that the 1.3% maximum desensitizer wax content will not be exceeded for the melt (see 4.5.5). Clean scrap produced in regular operations may be reused provided that it complies with the requirements for the grade of high explosive specified except for form. Scrap from floor sweepings, ventilation, and suction apparatus (other than from drilling operations specifically controlled to prevent contamination of explosives) shall not be used (see 6.5).

3.2 Assembled projectile. The assembled projectile shall comply with all requirements specified on Drawing (dwg) 8837902, and with all requirements specified in the applicable specifications.

3.3 Cavitation in high explosive charge

3.3.1 Cracks and projected cavities. Cracks and cavities within the explosive charge shall not exceed the requirements specified in Table I when determined in accordance with 4.5.1.1. Cavities having a maximum (max) projected length of 1/32 inch (in.) or less shall be disregarded.

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TABLE I

(See Figure 1 when loading deep cavity. See Figure 3 when loading shallow cavity).

Use Column D when loading deep cavity projectiles. Use Column D when loading shallow cavity projectiles.

	SEGMENTS				
	A	B	C	D	D'
Sum of projected areas of the cavities, excluding pipes, cracks and annular rings, square inch (sq. in.)	*1/64	1/4	1/2	1/2	1/2
Projected length of any cavity excluding pipes, cracks, and annular rings (in.)	1/8	1/2	1/2	3/4	3/4
Piping cavities, max. projected area (sq. in.)	0	1/4	1/2	-	1/2
Piping cavities, max. projected width (in.)	0	1/4	1/4	-	1/4
Cracks, max. projected width (in.)	0	1/32	1/32	-	-
Annular rings, max. projected width (in.)	0	0	1/4	-	1/4

*If the length of the largest cavity is 1/16 inch or less, the maximum total projected area may be 1/20 square inch. Piping cavitation is defined as cavitation located on or near the vertical center line of the projectile with the longitudinal axis (length) three times or greater than the width.

An Annular ring is defined as cavitation that takes the form of a ring or portion thereof around the periphery of the cast.

3.3.2 Surface cavitation. Cavities formed due to chipped or broken explosive from the sidewalls of the fuze well shall not extend in aggregate around more than 1/4 of the circumference. The

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max. dimension of any visible cavity in the faced surface of the charge shall not exceed 1/4 inch in any direction and the total area of all cavities shall not exceed 80 percent of the surface area. No individual point of explosive shall extend above the flat surface on which the liner would rest. (Cavities having a maximum dimension of 1/32 inch or less shall be disregarded in this computation as shall voids within the applicable forward region of the fuze well prescribed by drawing 8837902).

3.3.3 Porous areas. Porous areas shall be treated as cavities and shall be subject to the restrictions placed upon cavities, except that 80 percent of the projected length and 80 percent of the projected area shall be considered for acceptance purposes.

3.3.4 Cracked charges. No transverse crack shall be permitted in Segment A and not more than a total of two transverse cracks shall be permitted in Segments B and C combined. Cracks shall be disregarded in Segment D, and D'.

3.3.5 Base separation. There shall be no base separation greater than .015 inches between the base of projectile and the charge. This pertains to a sectioned projectile after the post cycle heat.

3.4 Loading of high explosive

3.4.1 Mating surfaces. The mating surfaces of threads shall be free of caked explosive and all threads shall be free of foreign material. If a chemical cleaner is required, acetone or methylene chloride shall be used. Care shall be taken to prevent the chemical cleaner from coming into contact with the charge and becoming entrapped in the projectile.

3.4.2 Solid explosives. Solid explosive shall not be in the projectile before loading or added to the projectile during initial pouring or cooling.

3.4.3 Coating. Projectile having the interior wall imperfectly coating shall not be loaded. The interior surface of the projectile shall be dry when the high explosive is poured into the projectile (see 4.4.2.3).

3.4.4 Projectile interior cavity. The interior cavity of the projectile prior to loading shall be free from foreign matter.

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3.4.5 Fuze well. The fuze well shall be free from any foreign matter or loose explosive prior to assembly of the liner.

3.4.6 Unsolidified pool of paint. There shall be no unsolidified pool of paint or blister in the bottom of the cavity prior to loading of the explosive.

3.4.7 Exterior surfaces. There shall be no explosive on the exterior surface of the projectile after loading.

3.4.8 Body cavity. The body cavity shall be free of scale, fins, burrs, draw marks, laminations, imbedded foreign matter, pits and sharp edges.

3.5 Proving ground test. The projectiles shall comply with the following requirements when tested as specified in 4.5.2.

3.5.1 Horizontal range

3.5.1.1 The mean range of any group of projectiles shall vary not more than ± 1% from the expected firing table range after correction to standard conditions.

3.5.1.2 The horizontal range probable error shall be no greater than .25 percent of range after correction for velocity variation.

3.5.2 Deflection

3.5.2.1 The mean deflection of any group of projectiles shall vary not more than ± 10% from the expected firing table deflection after correction to standard conditions.

3.5.2.2 The probable error in deflection shall be no greater than .06 percent of mean range.

3.5.3 Safety. There shall be no premature burst or indication of metal parts failure in the gun bore or in flight before impact, when the loaded projectile is tested between 54,000 & 58,000 pounds per square inch (psi) chamber pressure.

3.6 First Article inspection. This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

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3.7 Exudation. After heat cycling of TNT and Comp B filled projectiles, there shall be no exudation in the fuze well or exterior of the projectile.

3.8 Workmanship. The requirements of workmanship are as shown on the applicable drawing, in the applicable specifications and the following:

3.8.1 Cleaning. If required, the cleaning method shall not be injurious to any of the parts nor shall the parts be contaminated by the cleaning agent.

3.8.2 Protection of the rotating band. Care shall be exercised at all times so that the rotating band of each projectile will not be nicked or burred.

3.8.3 Painting. Painting shall comply with the requirements of the applicable drawing. All paint shall be dry to the touch before testing and packing for shipment. Drying time prior to testing shall be in accordance with the applicable specification.

3.8.4 Foreign matter. No part or assembly shall contain dirt, grease, chips, rust corrosion or other foreign matter, prior to inspection and shipping.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48076 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspections. The following types of inspections shall be performed on this item:

a. First Article Inspection
b. Quality Conformance Inspection

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4.3 First Article Inspection

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated.

Part Description	Drawing	Quantity
Projectile	8837902	20

4.3.2 Inspections to be performed. See MIL-A-48078 and Table II specified herein.

4.3.3 Rejection. See MIL-A-48078

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TABLE I - FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE			SHEET 1 OF 1	DRAWING NUMBER
	Projectile, 175mm, HE, M437 A1 or A2				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE - INSPECTION METHOD
	Projectile (Dwg. 8837902) Examination for Defects	20		3.2	4.4.2.1 to 4.4.2.5

NOTES

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4.4 QUALITY CONFORMANCE INSPECTION

4.4.1 Inspection lot formation. Inspection lots shall conform with the lot formation provisions of MIL-A-48078. In addition, each inspection lot of loaded projectiles shall contain:

a. Projectile metal parts of one lot interfix number from one manufacturer.
b. Loaded projectile of one weight zone.
c. Fuze well liners from one manufacturer produced in accordance with the same drawing, same drawing revision, same specification, and same specification revision.
d. TNT of the same type from one lot interfix number from one manufacturer, or Composition B of the same grade from one lot interfix number from one manufacturer which contains TNT from not more than one lot interfix number from one manufacturer.
e. Supplementary charges from one manufacturer produced in accordance with the same drawing, same drawing revision, same specification and same specification revision if loading deep cavity projectiles.

4.4.2 Examination. See MIL-A-48078.

a. Sampling plans. Unless otherwise specified in the Classification of Defects and Test tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE				DRAWING NUMBER
4.1.2.1	Projectile Prior to Loading		SHEET 1 OF 1		8837902
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE INSPECTION METHOD
Critical					
1	Foreign matter in projectile cavity		100%	3.4.4	Visual
2	Unsolidified pool of paint or blister in projectile cavity		100%	3.4.6	Visual/Manual
3	Interior metal defects		100%	3.4.8	Visual
Major					
101	Interior protective coating with bare spot		0.40%	3.4.3	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.8	Visual

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-45455E (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.2	Projectile and Liner (Prior to Assembly of Liner)				8797092/9258173
					NEXT HIGHER ASSEMBLY
					8837902
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE / INSPECTION METHOD
Critical	None defined				
Major					
101	Depth of fuze well, maximum		0.40%	3.2	Gage
102	Foreign matter in fuze well		0.40%	3.4.5	Visual
Minor					
201	Silicone compound not 360 degrees around liner or bottom (exterior surface only)		100%	3.2	Visual
202	Diameter of fuze well, maximum		0.65%	3.2	Gage
203	Cavitation in base of fuze well		0.65%	3.3.2	Visual
204	Fuze well sidewall cavitation excessive		0.65%	3.3.2	Visual
205	Presence of explosive on exterior of shell body		0.65%	3.4.7	Visual
206	Evidence of poor workmanship		0.65%	3.8	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-45455E (AR)

PARAGRAPH	TITLE	SHEET 1 OF 2	DRAWING NUMBER
4.4.2.3	Projectile, Prior to Assembly of Lifting Plug		8797092/9258173
			NEXT HIGHER ASSEMBLY
			8837902

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Liner missing		100%	3.2	Visual
2	Supplementary charge pad or supplementary charge missing (if applicable)		100%	3.2	Visual
Major					
101	Concentricity of inside diameter of liner with pitch diameter of thread including perpendicularity with face of body		0.40%	3.2	Gage
102	Depth to bottom of liner, min.		0.40%	3.2	Gage
103	Supplementary charge inverted (if applicable)		0.40%	3.2	Visual
104	Supplementary charge not removable by hand (if applicable)		0.40%	3.2	Visual/Manual
Minor					
201	Liner damaged		0.65%	3.2	Visual
202	Supplementary charge damaged (if applicable)		0.65%	3.2	Visual
203	Foreign matter in liner cavity including explosive		0.65%	3.2	Visual
204	Explosive in thread		0.65%	3.4.1	Visual

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-45455E (AR)

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
4.4.2.3	CONTINUATION SHEET				8797092/9258173
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	8837902 / PARAGRAPH REFERENCE / INSPECTION METHOD
Minor					
205	Liner removal by hand		0.65%	3.2	Manual
206	Spacer missing or damaged (if applicable)		0.65%	3.2	Visual
207	Silicone compound missing from threads		0.65%	3.2	Visual
208	Evidence of poor workmanship		0.65%	3.8	Visual

NOTES

DRUSR-QA Form 260 Jul 77 Replaces SAPPA-QA Form 2557 Feb 74 Which Is Obsolete

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.4	Projectile Assembly		8837902
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Projectile zone weight incorrect		100%	3.2	Balance
2	Zone stenciling incorrect		100%	3.2	Visual
3	Proving Ground Safety	5		3.5.3	4.5.2.1
Major					
101	Diameter of bourrelet, max (applies if bourrelet has been retouched)		100%	3.2	Gage
102	Torque test failure of lifting plug		0.40%	3.2	Test
103	Lifting plug gasket missing or improperly assembled		0.40%	3.2	Visual
104	Any weight zone punch mark inadequate		0.40%	3.2	Visual/Manual
105	Obturator web not cut between holes		0.40%	3.2	Visual
Minor					
201	Grommet missing, damaged or improperly assembled		0.65%	3.2	Visual
202	Rotating band damaged		0.65%	3.8.2	Visual
203	Protective coating with total damaged areas in excess of 1/4 square inch		0.65%	3.8.3	Visual
204	Marking misleading or unidentifiable (except for zones), including incorrect color code		0.65%	3.2	Visual
205	Evidence of poor workmanship		0.65%	3.8	Visual

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-45455E (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.5	Pallet				8857344
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Pallet damaged or improperly assembled		0.40%	3.2	Visual
102	DOD or NSN symbol missing or unidentifiable		0.40%	3.2	Visual
103	Projectile of different types on the same pallet		0.40%	3.2	Visual
104	Strapping or staple missing or loose		0.40%	3.2	Manual
Minor					
201	Number of projectile assemblies incorrect		0.65%	3.2	Visual
202	Marking misleading or unidentifiable		0.65%	3.2	Visual
203	Strapping or staple improperly engaged		0.65%	3.2	Visual

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4.4.3 Testing

4.4.3.1 Cavitation - (see 3.3 except 3.3.2). Cavitation defects are classified as follows:

Critical

1. Cavitation, cracks or annular rings in Segment A or B in excess of that permitted in Table I.

2. Cavitation or annular rings in Segment C in excess of that permitted in Table I.

3. Cracks in Segment C in excess of that permitted in Table I.

4. Cracks in excess of the number permitted by 3.3.4.

5. Base separation (see 4.4.3.1.1).

6. Cavitation in Segment D in excess of that permitted in Table I.

7. Cavitation in Segment D in excess of that permitted in Table I.

This test shall be performed on all projectiles (100 percent inspection). The radiographic test shall be performed as specified in 4.5.1, using equipment in accordance with 4.4.4.

4.4.3.1.1 Base separation - Critical defect. Projectiles showing the largest base separation as determined by radiographic examination prior to post cycle heat treatment shall be selected for sectioning at the following rate. Select projectiles at a rate of one (1) out of five hundred (500) projectiles to be included in the post cycle heat treatment lot with a minimum of two (2) samples per lot. These projectiles are to be included in the post cycle heat treatment and sectioned for determination of base separation. Thereafter, if insufficient samples are available with base separation, a sufficient number of samples shall be randomly selected from the post cycle heat treatment lot to make up the required sample. The samples selected for sectioning must be placed

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in the coolest area of the post cycle igloo. Any projectile after sectioning which has base separation exceeding .015 of an inch shall be classed defective, and the entire post cycle heat treatment lot rejected.

4.4.3.2 Proving ground test. The projectiles shall be observed for compliance with the requirements specified in 3.5. The test shall be performed as specified in 4.5.2.

Defects shall be classed as follows:

Category	Defects
Critical:	
1.	Premature burst or metal parts separation.
Major:	
101.	Mean range shall not vary more than ± 1% (see 3.5.1).
102.	Horizontal range probable error shall be no greater than .25 percent (see 3.5.1).
103.	Deflection shall not vary by more than ± 10% (see 3.5.2)
104.	Deflection probable error shall be no greater than .06% (see 3.5.2).
Minor:	None defined.

4.4.3.2.1 Functioning. The functioning test shall be conducted in accordance with 4.5.2.

4.4.3.2.1.1 Beginning with the first lot produced and continuing until three consecutive lots have complied with the acceptance criteria specified herein, a random sample of 15 projectiles shall be selected from each lot for this test. If a premature burst occurs, if the probable error of range or deflection of the sample multiplied by the factor 0.69 exceeds the applicable lot requirement or if the mean deviation from the expected range or deflection exceeds the maximum requirement, the lot shall be rejected.

4.4.3.2.1.2 After three consecutive lots have met the criteria of 4.4.3.2.1.1, a random sample of 10 projectiles shall be selected

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from each lot for this test. If a premature burst occurs, if the probable error of range or deflection of the sample multiplied by the factor 0.65 exceeds the applicable lot requirements or if the mean deviation from the expected range or deflection exceeds the maximum requirement, the lot shall be rejected.

4.4.3.2.2 Safety - (See 3.5.3) - Critical defect. Five projectiles shall be randomly selected from the lot for testing as specified in 4.5.2. If any projectile fails to comply with the applicable requirement, the lot shall be rejected.

4.4.3.2.3 Test validity. If for any reason the proving ground considers that the test conditions have detrimentally affected the test results, the test shall be considered invalid and a new test shall be performed with additional samples.

4.4.4 Inspection equipment. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and paragraph 6.3 herein.

4.4.5 Exudation - Minor defect. Six projectiles from the hottest area of the post cycle oven shall be used for the sample. Any projectiles with exudation shall be cause for rejection of the post cycle treatment lot.

4.5 Test Methods and Procedures

4.5.1 Radiographic examination

4.5.1.1 Determination of projectile cavitation. The projectiles shall be subjected to radiographic examination. The radiographic beam shall be located in a plane which is perpendicular to the longitudinal axis of the projectile and tangent to the base of its internal cavity. The outline of the explosive charge shall be clearly defined at all points. Examination of the radiograph shall be made to determine compliance with applicable requirements. No images of lead masks, containers, fixtures, identifying marks, or other extraneous features should project into the region of explosive under inspection, with the exception of specified penetrameters and radiation screening required to reduce film densities to readable levels in the thinner areas of the shell. Such screening should be free of any radiographically observable features in the projected area under inspection. Radiographic equipment,

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operations and procedures shall be qualified in accordance with MIL-STD-453 (see 4.5.1.2). In addition, the sensitivity of the radiographic technique shall be 2.0 percent and the minimum photographic density of the film used shall be no less than 1.25 at every point within the projected area of explosive, including the base, with the maximum density no greater than to permit adequate inspection over the entire area of interest, as determined in accordance with 4.5.1.3 and 4.5.1.4. All defect-like features in the area of interest on a radiographic image shall be treated as defects unless identified as extraneous artifacts or metal parts irregularities. In the event an original radiograph leaves doubt as to a shell's acceptability because of difficulty in evaluating the size, location, or classification of a defect, the projectile shall be rotated 90 degrees about its longitudinal axis from the original exposure and additional radiographs taken for clarification. Acceptability in all cases shall be judged on the basis of the film giving the largest projection, except in the case where a re-radiographic reveals multiple superimposed cavities, each of which individually complies with the requirements. All additional radiographs of a projectile for defect conditions will be identified on the film as retakes and stored; traceable to the original.

4.5.1.2 Radiographic equipment, operations and procedures. Radiographic equipment, operations and procedures shall be qualified in accordance with Standard MIL-STD-453.

4.5.1.3 Check of sensitivity of radiographic technique. The sensitivity of radiographic technique shall be determined by means of penetrameters fabricated as specified herein. The penetrameters shall be placed on the radiographic source side of a projectile so that the 2T holes are at the point shown in Figure 2. Three penetrameters for each projectile shall be used. When radiographing a number of similar projectiles, the frequency of the sensitivity check shall be no less than one in each fifty radiographs, and on a rotating ring, the penetrameters shall be used at one station for each complete rotation of the ring.

The sensitivity of the radiographic technique shall be considered satisfactory when the image of the penetrameters, including the 2T holes, are perceptible in accordance with MIL-STD-453.

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The penetrameters shall be made of steel having an approximate specific gravity of 7.8. The penetrameters shall be located as shown on Figure 2. For further details on construction of the penetrameter, see MIL-STD-453.

4.5.1.4 Check of photographic density of film. The photographic density of the film shall be determined on the film in the region of minimum film density within the projected area of the explosive charge. Film density shall be checked each hour of operation or fraction thereof, for each film processor. Greater frequency shall be required for any period during which difficulty is encountered in establishing the required densities. Photographic densities shall be measured with a diffuse density, photoelectric densitometer providing direct display of numerical density units. Accuracy of densitometer shall be maintained by calibrating with a photographic step tablet each hour immediately prior to measurement on the radiographic film. Calibration shall be made at a step tablet density level within .5D of that being measured on the radiographic film. Each point of density measurement on the radiographs shall be identified by a circumscribed circle (made with a suitable pen or pencil) of no more than 1/4 inch in diameter. All photographic step tablets used in normal operation shall be checked for accuracy monthly by calibrating the densitometer with a reference photographic step tablet calibrated by National Bureau of Standards or approved by ARRADCOM. Reference tablet shall be kept available, but shall be protected from wear and damage and shall not be used in daily operations. Any photographic step tablet having severe surface damage or showing a deviation from previous calibrations by .05D or more shall be replaced with a new, unused step tablet.

4.5.1.5 Review of radiographic sensitivity and techniques (see 6.9). The sensitivity of the radiographic techniques shall be reviewed by submitting one radiograph of the standard and a copy of radiographic procedures prior to the start of initial production. These radiographs shall be taken and processed using the same techniques as will be used in production. In addition, one radiographic of each of the projectile standards shall be submitted for review once a month for the first three months of each contract and every three months thereafter.

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4.5.1.6 Radiographic film. Only a film type and vendor as listed herein has been tested and approved by ARRADCOM for use in radiographing this projectile. A substitute item shall not be used before prior testing and/or approval by ARRADCOM, Dover, New Jersey 07801.

Eastman Kodak Company
1187 Ridge Road, West
Rochester, New York 14650

Film Types: AA, T, or M

E. I. DuPont de Nemours and Company
Photo Products Department
Wilmington, Delaware 19898

Film Types: NDT-55, NDT-65 or NDT-70

4.5.2 Proving ground

4.5.2.1 Proving ground. This test shall be conducted at a Government proving ground in accordance with the applicable acceptance test procedure. The sample projectiles with dummy fuzes shall be fired with a Zone 3 M86 propelling charge conditioned to 145°F. The charge shall be adjusted to attain between 54,000 & 58,000 psi in a cannon with a minimum of 50 percent wear life remaining. The results are to be recorded for individual items. Camera coverage shall be provided to observe the projectile in flight in order to record any malfunctions (see 6.6). In addition, radar shall be used to observe the projectile in flight in order to record any possible malfunctions.

4.5.2.2 Functioning. The projectile for this test shall be fired at charge 3 with the fuzes set super quick to obtain a mean range of 20,000 ± 500 meters. For comparison, an equal number of projectiles from a calibrated reference lot (see 6.10) shall be fired. The weight, center of gravity and moment of inertia, transverse and axial, and static and dynamic unbalance shall be taken and recorded. The individual velocities, ranges, and deflection shall be obtained and recorded. The probable error and mean of range, and deflection shall be calculated and recorded. The mean of range and deflection by correcting for reference calibration correction (see 6.10).

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4.5.3 Determination of base separation. After heat treatment the samples from each post cycle heat treatment lot shall be cooled to ambient and sectioned in a manner to expose the interface between the cast and the interior base of the projectile at the approximate center of the base. The projectile shall be positioned nose down or the cast shall be moved as far towards the nose with finger pressure as possible before measuring for base separation between the shell cavity and cast.

4.5.4 Exudation. The exterior of the projectile shall be visually inspected for exudation. If no exudate is found, remove the lifting plug and supplementary charge and inspect for exudation visually.

4.5.5 Determination of desensitizer content. Comp B riser scrap shall be batched in amounts appropriate to the production rate but not in excess of a 1 shift accumulation. For each batch of scrap, a sample of 4 randomly selected risers shall be crushed and blended. A 5 gram sample shall then be selected for determination of desensitizer content in accordance with MIL-C-401E. The amount of riser that can be added to virgin material shall then be calculated based on its desensitizer content, the desensitizer content of the virgin explosive as specified on the explosive data sheet and the total kettle weight.

5. PREPARATION FOR DELIVERY

5.1 Packing - Level A. The projectile shall be packed in accordance with Dwg. 8857344.

5.1 Marking. Marking shall be in accordance with Dwg. 8857344.

6. NOTES

6.1 Intended use. The components and assemblies covered by this specification are intended for use on the M437 Projectile.

6.2 Ordering data. See MIL-A-48078 (AR).

6.3 Submission of designs for approval - See 6.2.3 of MIL-A-48078. Submit equipment designs, as required to: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-QAR-I, Dover, New Jersey 07801.

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6.4 Photographic densities. "American Standards for Diffuse Transmission Density" published by American Standards Association, 70 East 40th Street, New York, has been satisfactory as a source of information on film sensities.

6.5 Clean explosive scrap. Clean explosive scrap is defined as scrap in kettles, pails, risers and loading machines used in the pouring and cooling operations and scrap formed during the drilling operation.

6.6 Proving ground. A copy of all proving ground data should be submitted to: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-QAR-Q and DRDAR-LCU-EP, Dover, New Jersey 07801.

6.7 Visual examination qualification. When compliance with the applicable requirement is in doubt as a result of visual examination, the characteristic may be measured or gaged to determine acceptability.

6.8 Radiographic procedures. Radiographic procedures standards and sample radiographs shall be submitted to the Commander, US Army Armament Research and Development Command, ATTN: DRDAR-QAR-Q, Dover, New Jersey 07801.

6.9 Drawings. Drawings listed in Section 2 of this specification under the heading US Army Armament Research and Development Command (ARRADCOM) may also include drawings prepared by and identified as, Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARRADCOM.

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6.10 Proving ground test. The reference projectiles shall be calibrated to establish a calibration range and deflection and to determine the departure of range and deflection from the firing table at charge 3 at a range of 20,000 meters when corrected to a standard velocity of 3000 fps and standard meterological conditions. The test projectile data shall be corrected to standard conditions by adding or subtracting the average departure of the reference projectiles from the firing table-reference calibration correction.

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Preparing Activity
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Project No.: 1320-A504

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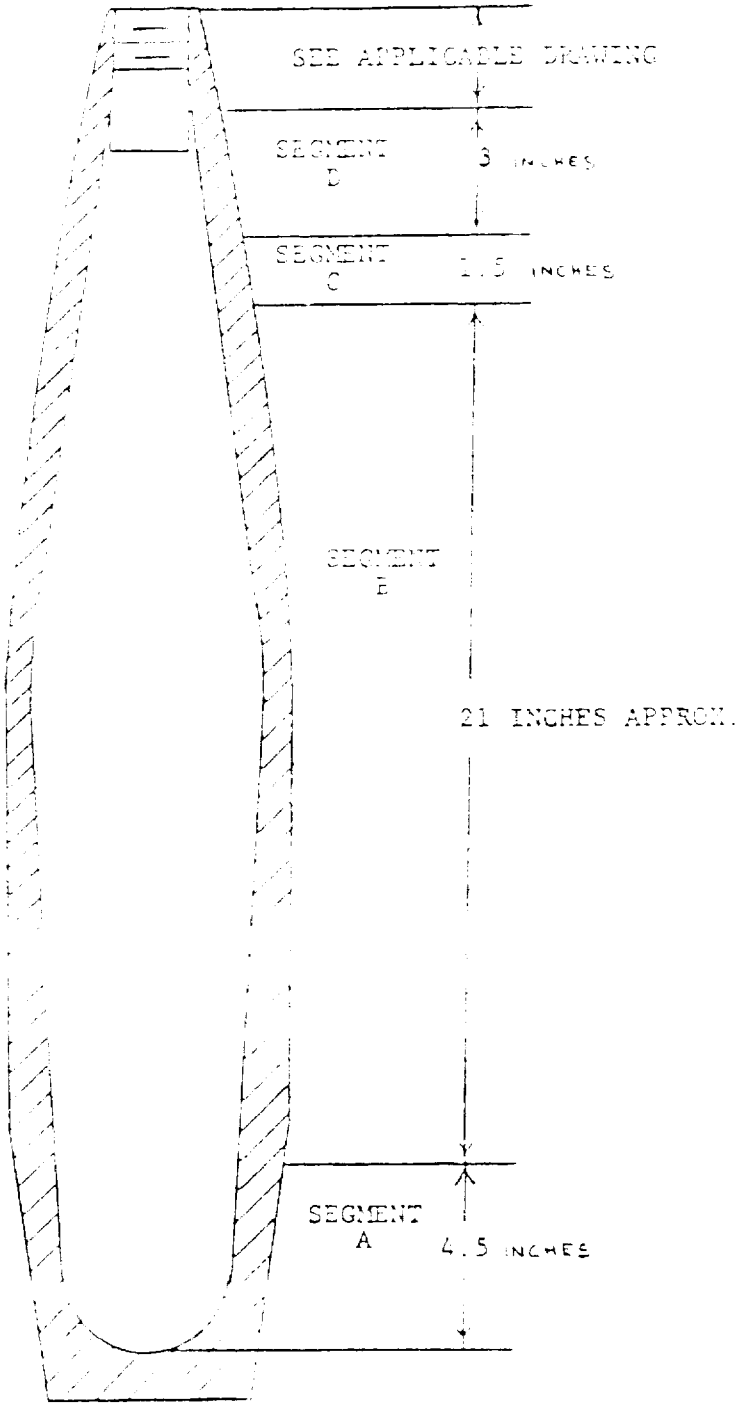


FIGURE 1

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Penetrameter Calculation
For 175MM, HE, M437A2

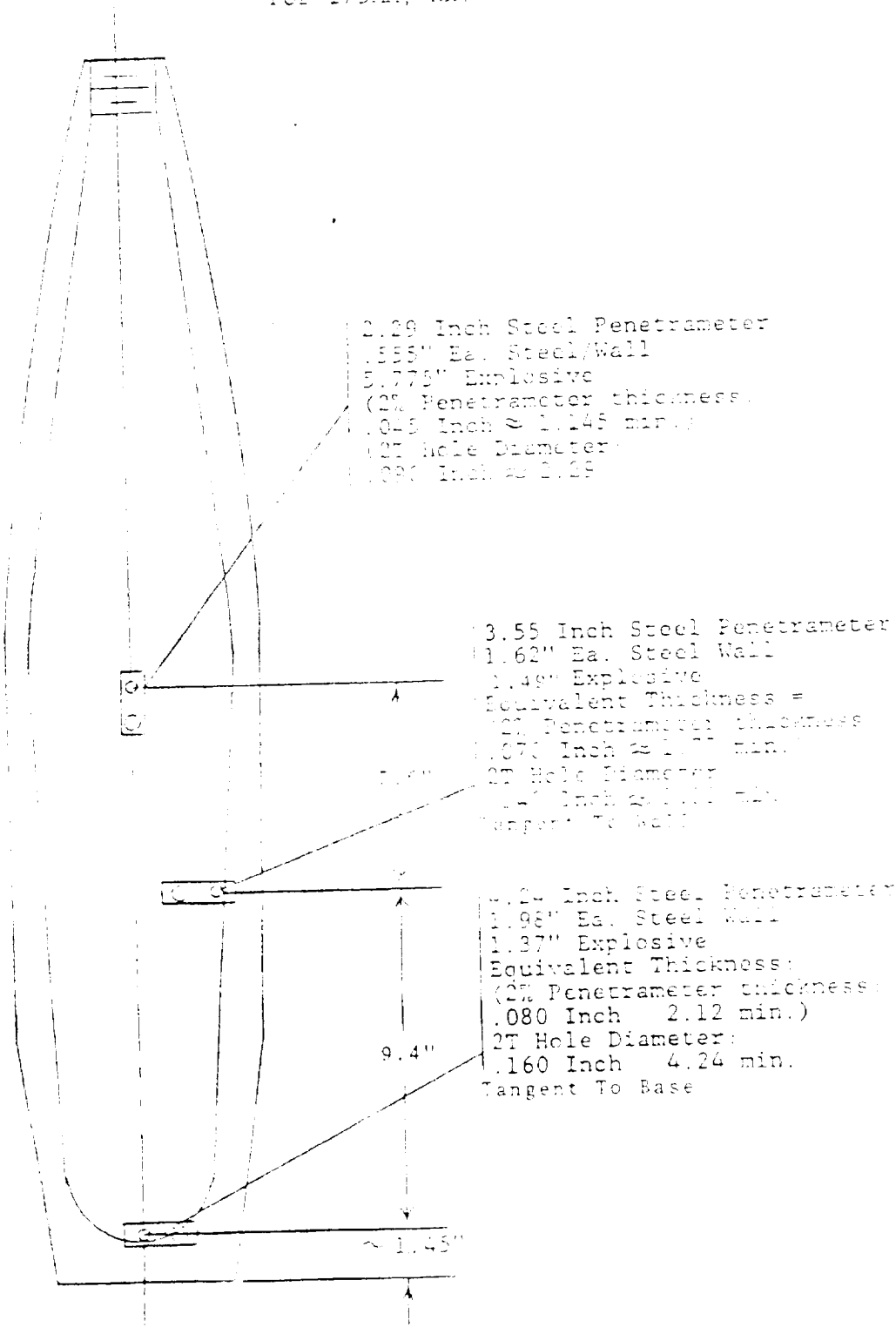
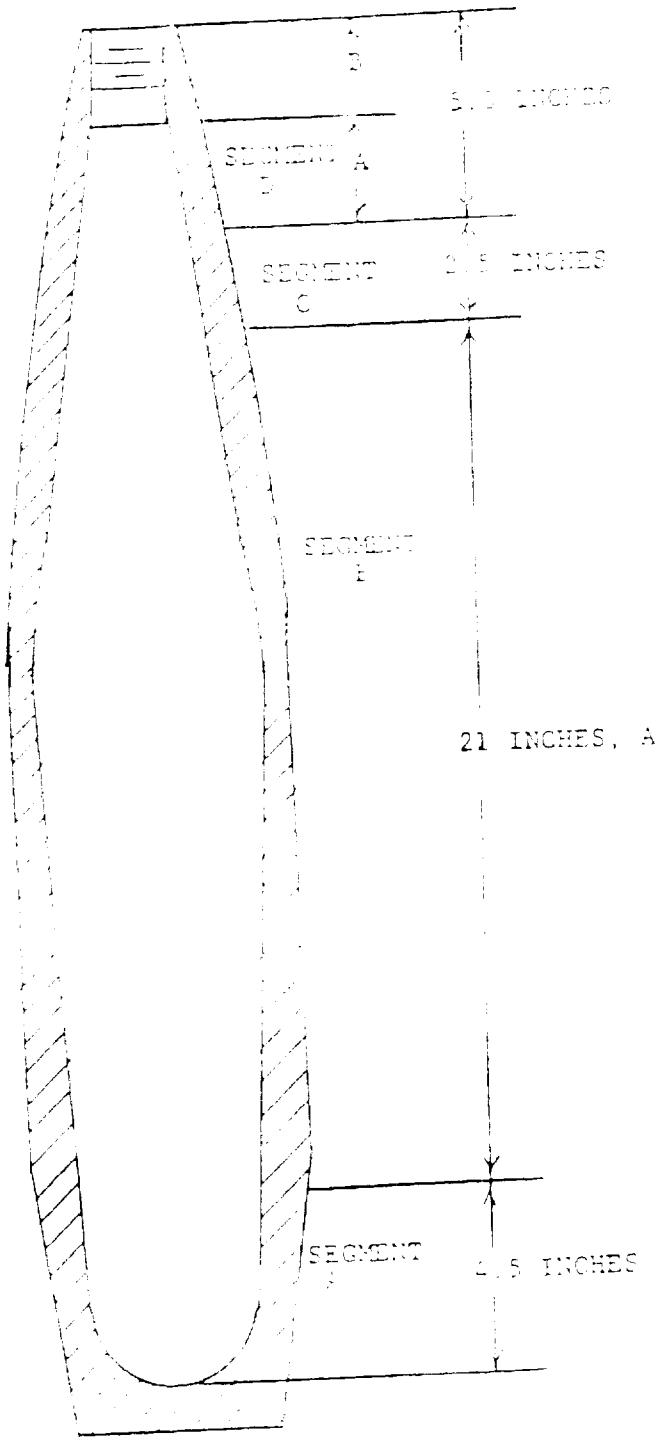


FIGURE 2

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A - Area from bottom of tube well face to five (5) inches from the nose of the shell.

B - Area from nose of shell to start of Segment D.

FIGURE 3

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